Mission Control Center

Mission_Control_Centers

Column Name	Data Type	Constraints	Description
mcc_id	INT	PRIMARY KEY	Unique identifier for each mission control center
center_name	VARCHAR	NOT NULL	Official name of the control center
location_city	VARCHAR		City where center is located
timezone	VARCHAR		Timezone of the facility

Missions

Column Name	Data Type	Constraints	Description
mission_id	INT	PRIMARY KEY	Unique identifier for each mission
mcc_id	INT	FOREIGN KEY REFERENCES Mission_Control_Centers(mcc_id)	Controlling center for this mission
mission_name	VARCHAR	NOT NULL	Name of the mission
mission_goal	TEXT		Description of mission objectives
start_date	DATE		When mission began

Satellite

Column Name	Data Type	Constraints	Description
sat_id	INT	PRIMARY KEY	Unique identifier for each satellite
mission_id	INT	FOREIGN KEY REFERENCES Missions(mission_id)	Mission this satellite belongs to
model_name	VARCHAR		Manufacturer model name/designation
status	VARCHAR		Current operational status

Satellite_State

Column Name	Data Type	Constraints	Description
state_id	INT	PRIMARY KEY	Unique state record identifier
sat_id	INT	FOREIGN KEY REFERENCES Satellites(sat_id), UNIQUE	Satellite this state belongs to (one-to-one)
orbit_id	INT		Current orbital level identifier
theta_deg	DECIMAL		Angular position in orbit (degrees)
altitude_km	DECIMAL		Current altitude above Earth
velocity_kms	DECIMAL		Current orbital velocity
last_updated	TIMESTAMP		When this state was recorded

Subsystems

Column Name	Data Type	Constraints	Description
sub_id	INT	PRIMARY KEY	Unique subsystem identifier
sat_id	INT	FOREIGN KEY REFERENCES Satellites(sat_id)	Parent satellite
sub_type	VARCHAR		Type of subsystem (e.g., Power, Propulsion)
mode	VARCHAR		Current operational mode

Components

Column Name	Data Type	Constraints	Description
comp_id	INT	PRIMARY KEY	Unique component identifier
sub_id	INT	FOREIGN KEY REFERENCES Subsystems(sub_id)	Parent subsystem
part_name	VARCHAR		Name/identifier of the part
health_index	INT		Current health percentage/indicator

Telemetry_Logs

Column Name	Data Type	Constraints	Description
log_id	BIGINT	PRIMARY KEY	Unique log entry identifier
comp_id	INT	FOREIGN KEY REFERENCES Components(comp_id)	Component providing telemetry
log_timestamp	TIMESTAMP		When reading was taken
val	DECIMAL		Reading value
is_eclipse	BOOLEAN		Whether satellite was in eclipse

Command_Vault

Column Name	Data Type	Constraints	Description
cmd_id	INT	PRIMARY KEY	Unique command identifier
hex_code	VARCHAR	UNIQUE, NOT NULL	Actual hex command (e.g., "0x5F")
description	TEXT		What this command does
power_cost_watts	DECIMAL		Power required to execute

Command_History

Column Name	Data Type	Constraints	Description
hist_id	INT	PRIMARY KEY	Unique history record identifier
sat_id	INT	FOREIGN KEY REFERENCES Satellites(sat_id)	Target satellite
cmd_id	INT	FOREIGN KEY REFERENCES Command_Vault(cmd_id)	Command sent
execution_time	TIMESTAMP		When command was executed

Maneuver_Plans

Column Name	Data Type	Constraints	Description
plan_id	INT	PRIMARY KEY	Unique plan identifier
sat_id	INT	FOREIGN KEY REFERENCES Satellites(sat_id)	Satellite to maneuver
target_orbit	INT		Destination orbit level
thrust_val	DECIMAL		Thrust value/amount
approval_status	VARCHAR		Current approval state
execution_time	TIMESTAMP		When to execute

Transaction_Logs

Column Name	Data Type	Constraints	Description
tx_id	INT	PRIMARY KEY	Unique transaction identifier
plan_id	INT	FOREIGN KEY REFERENCES Maneuver_Plans(plan_id)	Associated maneuver plan
pre_fuel_kg	DECIMAL		Fuel level before maneuver
pre_altitude_km	DECIMAL		Altitude before maneuver
tx_status	VARCHAR		Transaction status (pending/committed/rolled_back)

Space_Debris_Catalog

Column Name	Data Type	Constraints	Description
debris_id	INT	PRIMARY KEY	Unique debris identifier
cluster_id	INT		Group/cluster identifier
theta_deg	DECIMAL		Angular position in orbit
orbit_id	INT		Current orbital level
danger_radius_km	DECIMAL		Size/threat radius

Observations [Bridge Table]

Column Name	Data Type	Constraints	Description
obs_id	INT	PRIMARY KEY	Unique observation identifier
sat_id	INT	FOREIGN KEY REFERENCES Satellites(sat_id)	Observing satellite
debris_id	INT	FOREIGN KEY REFERENCES Space_Debris_Catalog(debris_id)	Observed debris
detection_timestamp	TIMESTAMP		When observation occurred
confidence_score	DECIMAL		Certainty of identification (0-100)

Celestial_Bodies

Column Name	Data Type	Constraints	Description
body_id	INT	PRIMARY KEY	Unique identifier
body_name	VARCHAR		Name (e.g., "Sun", "Earth")
fixed_theta	DECIMAL		Fixed angular position

Ground Station and Bridge Tables

Ground_Station

Column Name	Data Type	Constraints	Description
gs_id	INT	PRIMARY KEY	Unique ground station identifier
gs_location	VARCHAR		Physical location description

Department

Column Name	Data Type	Constraints	Description
dep_id	INT	PRIMARY KEY	Unique department identifier
gs_id	INT	FOREIGN KEY REFERENCES Ground_Station(gs_id)	Ground station this department belongs to
dep_name	VARCHAR		Department name

Personnel

Column Name	Data Type	Constraints	Description
person_id	INT	PRIMARY KEY	Unique personnel identifier
dep_id	INT	FOREIGN KEY REFERENCES Department(dep_id)	Department employee belongs to
person_name	VARCHAR		Full name of personnel
role	VARCHAR		Job role/function
is_working	BOOLEAN		Whether currently on shift

Dish

Column Name	Data Type	Constraints	Description
dish_id	INT	PRIMARY KEY	Unique dish identifier
gs_id	INT	FOREIGN KEY REFERENCES Ground_Station(gs_id)	Ground station housing this dish
elevation_angle	INT		Current elevation angle in degrees
max_distance	INT		Maximum communication range
is_transmitting	BOOLEAN		Whether dish is currently active

Access_Time [Bridge 1]

Column Name	Data Type	Constraints	Description
access_id	INT	PRIMARY KEY	Unique access window identifier
sat_id	INT	FOREIGN KEY REFERENCES Satellites(sat_id)	Satellite being contacted
gs_id	INT	FOREIGN KEY REFERENCES Ground_Station(gs_id)	Ground station for this pass
start_time	TIME		Start time of communication window
end_time	TIME		End time of communication window
date	DATE		Date of the pass
status	VARCHAR		Current status of this pass
is_succesful	BOOLEAN		Whether data was successfully exchanged

Pass_Assignment [Bridge 2]

Column Name	Data Type	Constraints	Description
assignment_id	INT	PRIMARY KEY	Unique assignment identifier
access_id	INT	FOREIGN KEY REFERENCES Access_Time(access_id)	The scheduled pass
dish_id	INT	FOREIGN KEY REFERENCES Dish(dish_id)	Dish assigned to this pass
actual_time	TIME		When transmission actually occurred
actual_date	DATE		Date of transmission

Command_Authorization [Bridge 3]

Column Name	Data Type	Constraints	Description
auth_id	INT	PRIMARY KEY	Unique authorization record
cmd_id	INT	FOREIGN KEY REFERENCES Command_Vault(cmd_id)	Command being authorized
person_id	INT	FOREIGN KEY REFERENCES Personnel(person_id)	Person authorizing the command
approval_time	TIMESTAMP		When approval was given

Index and Role Specification

Index Specification

Table	Column(s)	Index Name	Rationale
Telemetry_Logs	val	idx_telemetry_val	Find high/low values for alerting
Space_Debris_Catalog	orbit_id, risk_level	idx_debris_spatial_risk	Identify high-risk debris
Satellite_State	sat_id	idx_satellite_state	Fast JOIN operation
Access_Time	sat_id	idx_access_time_sat	Fast JOIN operation
Access_Time	gs_id	idx_access_time_gs	Fast JOIN operation
Dish	gs_id	idx_access_time_dish	Fast JOIN operation
Command_History	sat_id, execution_time	idx_command_history_sat_time	Command audit. A complete history
Observations	debris_id, detection_timestamp	idx_observations_debris_time	Track specific debris over time
Telemetry_Logs	comp_id, log_timestamp	idx_telemetry_component_time	Component health history
Satellite_State	sat_id	idx_satellite_state_sat_unique	[UNIQUE] One-to-one relationship with Satellites
Command_Vault	hex_code	idx_command_vault_hex_unique	[UNIQUE] Ensure no duplicate hex commands

Role Specification

Role Name	Description	Permissions
Mission Control Engineer [MCE]	Receives data from DA on a Dashboard. And then selects command to send to Ground Station depending on the next available time window	**Read Access:** Views (Dashboard), Satellite_State, Telemetry_Logs, Access_Time **Write Access:** Command_History, Maneuver_Plans **No Access:** Personnel tables, Department structure, Ground Station hardware config
Ground Station Operator [GSO]	Receives command from MCE, angles the dish to align with Satellite and transmits data	**Read Access:** Access_Time, Pass_Assignment, Dish, Satellites **Write Access:** Access_Time (status, is_successful), Dish (is_transmitting) **No access:** Command_Vault, Maneuver_Plans, Personnel
Data Analyst [DA]	Analyzes the Data at the MCC, uses view and join to create MCE dashboard	**Read Access:** All tables and is able to create views along with temp tables **Write Access:** Only on temp tables **No Access:** Complete flow of data